João André **Roque Costa**

Software, Data & AI Engineer
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Python and AI engineer with an MSc in Computer Science and Engineering, focused on generative-AI-adjacent work, Azure data services, and automation. Built Python prototypes for data migration, reporting, and monitoring with Azure AI Foundry, Databricks, Azure Data Factory, RAG, OCR, and SQL-backed systems.

Professional Experience

Tech Analyst — *Coimbra, Portugal*
Deloitte — *Apr. 2026 - Oct. 2026*

- Built a Python prototype for data migration and synthetic-data generation with Azure AI Foundry, Blob Storage, and MongoDB.
- Automated reporting and data extraction workflows with Python and Playwright, and contributed RAG and OCR designs for document analysis and structured data extraction.
- Delivered an Azure subscription migration for Databricks and Data Factory assets, coordinating dependent services across the Azure data platform.

Teaching Assistant — *Lisbon, Portugal*
Instituto Superior Técnico — *Oct. 2024 - Jul. 2025*

- Taught and evaluated practical classes in Databases and Data Science, adapting explanations to different audiences and tracking student progress.
- Guided Python work in data preparation, statistical analysis, and visualization with pandas and matplotlib, emphasizing reproducible analysis and clear communication of results.

Education

M.Sc. Computer Science and Engineering — *Lisbon, Portugal*
Instituto Superior Técnico — *Sep 2023 - Nov 2025*

- Focused on Artificial Intelligence and Data Science, including machine learning, cloud computing, multi-agent systems, and large-scale data analysis.
- Relevant Coursework: Data and Information Systems Management, Data Analysis and Integration, Cloud Computing and Virtualization, Machine Learning, Deep Learning, Autonomous Agents and Multi-Agent Systems.
- Thesis: Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)
- Thesis: Developed a decomposition-based extension to MVU for nonlinear dimensionality reduction on disconnected datasets.
- Thesis: Designed a parallelizable method for large datasets, preserving local geometry while producing comparable embeddings.

B.Sc. Computer Science and Engineering — *Lisbon, Portugal*
Instituto Superior Técnico — *Sep 2020 - Jul 2023*

- Relevant Areas: Algorithms, distributed systems, operating systems, database theory, C/C++, and SQL.
- Relevant Project: Applied C/C++ in systems projects and designed relational schemas and transactions in SQL.

Extracurricular

- Built a self-hosted Python monitoring service with historical trend analysis, alerts, and an operational dashboard for tracking system state.
- Developed an event-driven Discord matchmaking service using ELO ratings, PostgreSQL match history, and Redis queues and caching.
- Built automation and API integrations for multi-platform price monitoring, risk management, optimal stake sizing, and inventory tracking.

Skills & Competencies

Python and GenAI	Python, Azure AI Foundry, Playwright, RAG, OCR.
Cloud and Data Platforms	Azure, Databricks, Data Factory, Blob Storage.
Data and Databases	SQL, PostgreSQL, MongoDB, MySQL, Redis, data pipelines.
Operations	Monitoring, dashboards, alerts, Linux, Git.
Analysis	Data analysis, statistics, machine learning.
Collaboration	Cross-functional communication, mentoring, technical presentations.
Languages	Portuguese (Native), English (Fluent).